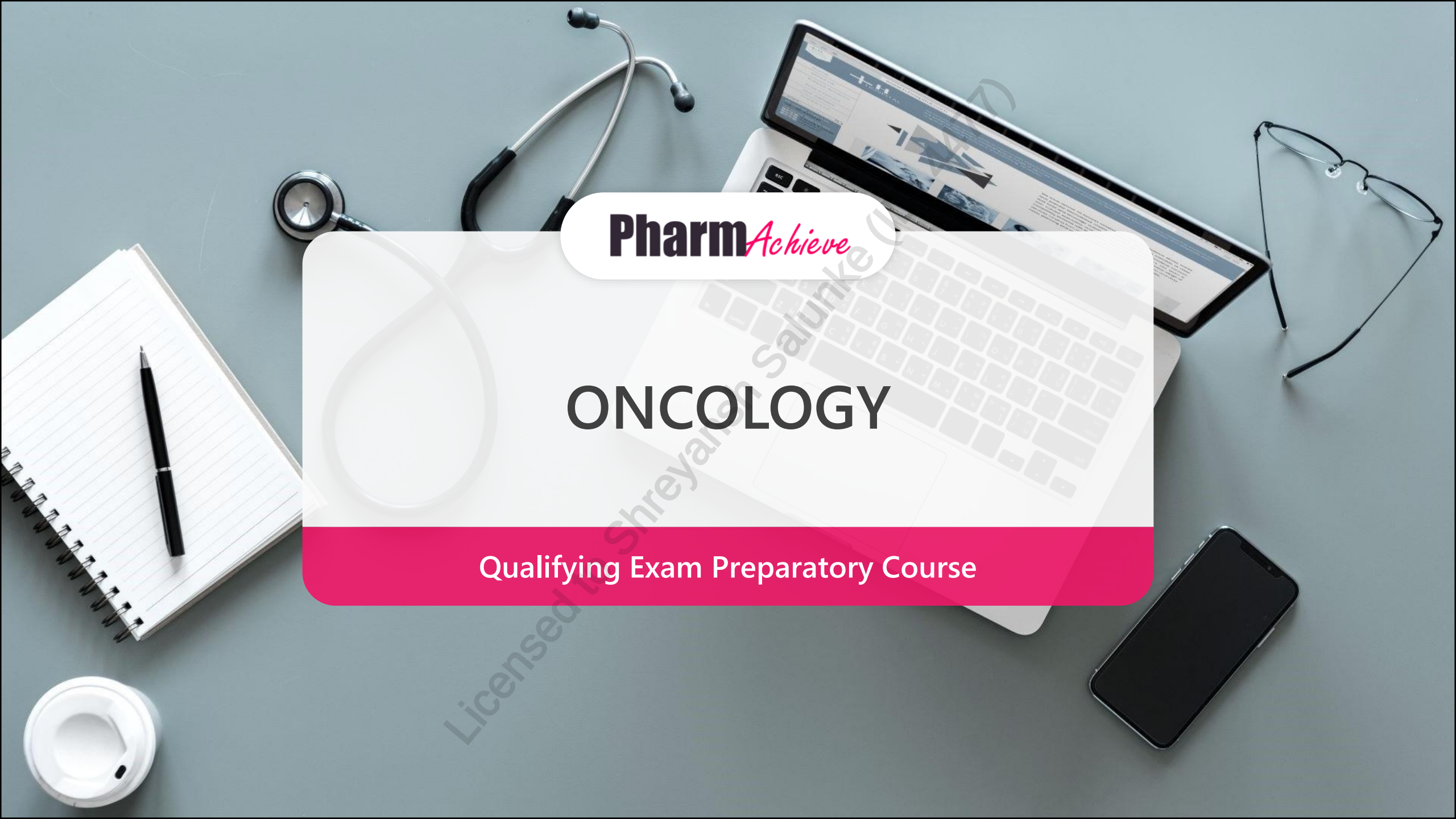


A top-down view of a grey desk with various medical and study items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill-shaped graphic is centered over the laptop.

Pharm*Achieve*

ONCOLOGY

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

5-HT	Serotonin	EF	Ejection Fraction
AC	Doxorubicin (Adriamycin) and Cyclophosphamide	EGFR	Epidermal Growth Factor Receptor
ACEi	Angiotensin Converting Enzyme Inhibitor	ESA	Erythropoiesis-Stimulating Agents
APR	Aprepitant	G-CSF	Granulocyte-Colony Stimulating Factor
BB	Beta-blocker	GI	Gastrointestinal
BM	Bowel Movement	GU	Genitourinary
BSA	Body Surface Area	Hb	Hemoglobin
BSE	Breast Self-Examination	IV	Intravenous
CAD	Coronary Artery Disease	LMWH	Low Molecular Weight Heparin
CCB	Calcium Channel Blocker	LHRH	Luteinizing Releasing Hormone
CINV	Chemotherapy-Induced Nausea/Vomiting	LV	Left Ventricle
CTZ	Chemoreceptor Trigger Zone	MCP	Metoclopramide
DEX	Dexamethasone	N/V	Nausea/Vomiting
DMSO	Dimethyl Sulfoxide	NK1	Neurokinin 1 receptor antagonist
DNA	Deoxyribonucleic acid	PSA	Prostate-Specific Antigen
DOAC	Direct Oral Anticoagulant	SERMs	Selective Estrogen Receptor Modulator
DOP	Dopamine antagonist	SNRI	Serotonin-Norepinephrine Reuptake Inhibitor
DRE	Digital Rectal Exam	TCA	Tricyclic Antidepressant
DVT	Deep Vein Thrombosis	VTE	Venous Thromboembolism
ECOG	Eastern Cooperative Oncology Group		

DEFINITIONS

Cancer (malignancy): Invasion into normal tissues, bone, and lymph nodes, causing **uncontrolled proliferation and spread of abnormal cells**

Hematological: lymphoma, leukemia

Solid tumors: breast, lung, colorectal, prostate

DIAGNOSIS AND STAGING SOLID TUMORS

DIAGNOSIS

Tissue biopsy and histology for diagnosis and to determine treatment and prognosis

STAGING

Classify extent of disease to determine treatment and prognosis

Staging:

TNM classification system

T: size of the primary **tumour**

N: regional lymph **node** metastasis

M: distant **metastasis**

TNM STAGING*

Category	Meaning
T (tumour)	• TX – main tumor cannot be measured • T0 – no evidence of primary tumor • T1, T2, T3, T4 – the higher the number, the larger the tumor size
N (lymph node involvement)	• NX – regional lymph nodes cannot be assessed • N0 – no cancer detected in regional lymph nodes • N1, N2, N3 – the higher the number, the more lymph nodes with cancer
M (distant metastasis)	• MX – distant metastasis cannot be measured • M0 – no evidence in distant metastasis • M1 – there is distant metastasis

*A higher number indicates an increase in tumour size and spread

RISK FACTORS

- Smoking
- Alcohol
- UV radiation
- Ionizing radiation
- Viruses
- Occupational exposure
- Obesity
- Sedentary lifestyle
- Diet
- Drugs
- Genetics

SCREENING

- **Breast:** breast self-examination (BSE), clinical breast examination, mammography
- **Colorectal:** colonoscopy, stool DNA test
- **Cervical:** pap smear
- **Prostate:** digital rectal exam (DRE), prostate-specific antigen (PSA) blood test

CLINICAL PRESENTATION

- **Variable:** depends on type of cancer, but may include weight loss, fatigue, anorexia, change in bowel habits, night sweats

GOALS OF THERAPY

- ✓ Cure disease
- ✓ Prevent recurrence
- ✓ Slow disease progression
- ✓ Relieve symptoms
- ✓ Improve quality of life
- ✓ Prolong life

TREATMENT

Surgery

- Prevention
- Removal of cancer

Pharmacological options

- Chemotherapy - Adjuvant vs neoadjuvant vs palliative
- Hormonal (endocrine) treatment
- Targeted therapy
- Immunotherapy

Radiation

- Eradicate localized sensitive tumors
- "Radiation recall:" skin redness, swelling, and peeling of radiation site after chemotherapy

CHEMOTHERAPY

DNA Damage → ↓ DNA Synthesis → ↓ Cell Division

Mechanism of action

- **Targets cell division of rapidly growing cells** (i.e. cancerous cells) and interferes with cell division
- **Toxic to rapidly growing cells**

Considerations when choosing chemotherapy agent:

- Type of cancer
- Stage of disease (prognosis)
- Patient performance status (ECOG/Karnofsky)
- Comorbid disease states
- Patient preference

Combination therapy:

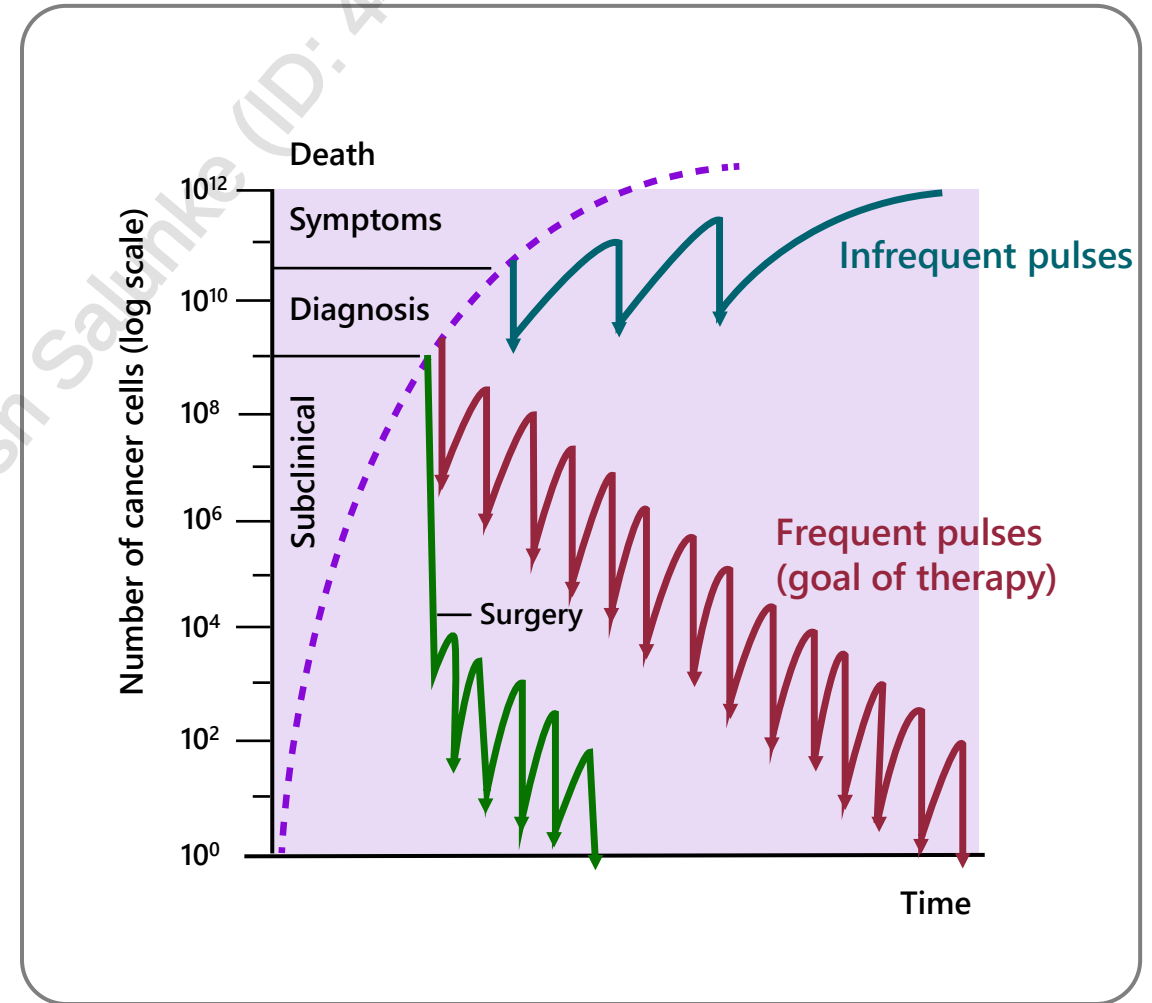
- **Include multiple agents to ensure broad coverage against resistant cancer cells**
- Include agents with:
 - Evidence in combination for specific cancer type
 - Optimal dose and schedule
 - Demonstratable single-agent activity
 - Different mechanisms of action
 - Overlapping toxicities may present

CHEMOTHERAPY

LOG KILL HYPOTHESIS

Cytotoxic Treatments:

Doses are given in frequent pulses to reduce cancer cells gradually until **almost zero**



TYPES OF ANTI-CANCER DRUGS - EXAMPLES

Chemotherapy



- Alkylating agents
- Epipodophyllotoxins
- Taxanes
- Antimetabolites
- Antitumor antibiotics
- Camptothecins
- Vinca alkaloids

Hormonal (endocrine) therapy

- Aromatase inhibitors
- Selective estrogen receptor modulators (SERMs)
- Anti-androgen medications
- LHRH (luteinizing hormone releasing hormone) agonists

Targeted therapy

- Anti-angiogenesis medications
- Apoptosis inducing drugs
- Tyrosine kinase inhibitors

Immunotherapy

- Biologic response modifiers
- Immune checkpoint inhibitors
- Immunomodulatory drugs

ALKYLATING AGENTS

CHEMOTHERAPY AGENT

Mechanism of action

- Covalently binds highly reactive alkyl groups to DNA strands – results in faulty DNA being made
- Cross-links DNA to prevent unwinding – results in DNA breakage in tumor cells
- Causes mispairing of nucleotides → leading to mutations to DNA

Common side effects

- GI adverse effects - nausea/vomiting
- Myelosuppression
- Mucositis

Examples

- Busulfan, cyclophosphamide, ifosfamide, melphalan, chlorambucil, carmustine, dacarbazine
- Platinum compounds: cisplatin, carboplatin, oxaliplatin

EPIPODOPHYLLOTOXINS

CHEMOTHERAPY AGENT

Mechanism of action	• Inhibits topoisomerase II • Enzyme responsible for unwinding double-strand DNA • Causes single and double-strand DNA breaks • Decreases DNA synthesis
Common side effects	• Nausea/vomiting (less common) • Myelosuppression • Mucositis • Alopecia
Examples	• Etoposide

TAXANES

CHEMOTHERAPY AGENT

Mechanism of action

- Promotes assembly and stabilization of microtubules → apoptosis
- **Inhibits cell replication and angiogenesis**

Common side effects

- Nausea/vomiting
- Myelosuppression
- Mucositis
- Fluid retention
- Alopecia
- Neuropathy/pain
- Hypersensitivity reactions

Examples

- Docetaxel
- Paclitaxel

ANTIMETABOLITES

CHEMOTHERAPY AGENT

Mechanism of action

- Reassembles nucleotides and incorporates into DNA/RNA
- **Decreases DNA/protein synthesis**
- Includes: pyrimidines, purines, and folate antagonists

Common side effects

- Diarrhea
- Myelosuppression
- Mucositis
- Skin irritation
- Heartburn/anorexia

Examples

- 5-fluorouracil
- Cytarabine
- Methotrexate
- Fludarabine
- Mercaptopurine
- Capecitabine

ANTITUMOR ANTIBIOTICS

CHEMOTHERAPY AGENT

Mechanism of action	• Inserts between DNA base pairs and inhibits DNA and RNA synthesis • Uncoils DNA helix • Induces DNA strand breakage by free radical formation
Common side effects	• Diarrhea • Myelosuppression • Tissue necrosis • Nausea/vomiting • Anorexia • Mucositis • Cardiotoxicity (anthracyclines)
Examples	• Anthracyclines: daunorubicin, doxorubicin, epirubicin, idarubicin • Bleomycin, dactinomycin, mitomycin

CAMPTOTHECINS

CHEMOTHERAPY AGENT

Mechanism of action	• Binds topoisomerase I • Enzyme responsible for single-strand breaks • Decreases DNA synthesis by inhibiting replication leading to cell death
Common side effects	• Diarrhea • Myelosuppression • Fatigue • Fever • Nausea/vomiting • Abdominal pain • Alopecia
Examples	• Irinotecan • Topotecan

VINCA ALKALOIDS

CHEMOTHERAPY AGENT

Mechanism of action

- Prevents polymerization of tubulin to form microtubules
- Induces depolymerization of formed tubules
- **Inhibits mitosis**

Common side effects

- Neurotoxicity
- Dizziness and vertigo
- Myelosuppression
- Skin necrosis (extravasation)
- Mucositis
- Constipation
- Alopecia)

Examples

- Vincristine
- Vinblastine
- Vinorelbine

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- Tyrosine kinase inhibitors

Immunotherapy

- Biologic response modifiers
- Immune checkpoint inhibitors
- Immunomodulatory drugs

ANTI-ANGIOGENESIS DRUGS

TARGETED THERAPY

Mechanism of action

- **Blocks the signals inside cancer cells that trigger blood vessel growth**

Common side effects

- Bleeding
- Clots
- Hypertension
- Proteinuria

Examples

- Bevacizumab (Avastin®)
- Some tyrosine kinase inhibitors (e.g. axitinib)

APOPTOSIS-INDUCING DRUGS

TARGETED THERAPY

Mechanism of action

- Stimulates cells that have lost the ability to undergo apoptosis to regain their sensitivity
- **Increases sensitivity to other chemotherapy agents**

Common side effects

- Nausea/vomiting
- Diarrhea
- Constipation
- Insomnia
- Muscle or bone pain
- Rash
- Skin irritation
- Pancytopenia
- Tumor Lysis Syndrome

Examples

- Bortezomib (Velcade®): *proteasome inhibitor*

TYROSINE KINASE INHIBITORS

TARGETED THERAPY

Mechanism of action	• Inhibits the following, resulting in cell death: • Tyrosine kinase receptors • Platelet-derived growth factor receptors • Vascular endothelial growth factor receptors	
Common side effects	• Myelosuppression • Fatigue • Nausea/vomiting	• Diarrhea • Mucositis/stomatitis • Skin reactions
Examples	• Sunitinib • Sorafenib	• Pazopanib • Imatinib



Mucositis that affects speech is a red flag that warrants urgent referral to the Emergency Department

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AROMATASE INHIBITORS

ENDOCRINE THERAPY

Mechanism of action

- Inhibits aromatase enzyme, which catalyzes the final conversion of androgens to estrogens in peripheral tissues, including breast tissue

Common side effects

- Hot flashes
- Fatigue
- Arthralgia/myalgia
- Headache
- Peripheral edema
- Hypercholesterolemia
- Osteoporosis

Examples

- Anastrozole
- Letrozole
- Exemestane

SELECTIVE ESTROGEN RECEPTOR MODULATOR

ENDOCRINE THERAPY

Mechanism of action	- Competitively binds to certain estrogen receptors on tumors to inhibit estrogen effects – antagonist effects on breast - Reduces DNA synthesis by stalling cell cycle - Estrogen agonist on endometrial and bone tissue
Common side effects	- Hot flashes - Thromboembolic events - Weight loss - Hypertension - Amenorrhea - Mood changes - Vaginal discharge
Examples	Tamoxifen

LHRH AGONISTS

ENDOCRINE THERAPY

Mechanism of action

- Downregulates LHRH receptors with chronic administration
- Inhibits secretion of luteinizing hormone
- **Decreases androgen and growth of androgen-dependent prostate cancer**
- **Reduces ovarian secretion of estradiol and progesterone in women for treatment of estrogen-dependent cancers**

Common side effects

- Decreased libido
- Hot flashes
- Gynecomastia
- Anemia
- Joint pain
- Erectile dysfunction

Examples

- Buserelin
- Goserelin
- Leuprolide

ANTI-ANDROGENS

ENDOCRINE THERAPY

Mechanism of action	• Inhibits binding of androgens to androgen receptors (androgen receptor blocker) • Newer agents act at multiple steps in the androgen receptor signaling pathway	
Common side effects	• Decreased libido • Impotence • Erectile dysfunction • Testicular atrophy • Hot flashes	• Nausea • Diarrhea • Fatigue • Gynecomastia
Examples	• Flutamide • Bicalutamide	• Enzalutamide • Apalutamide

GLUCOCORTICOIDS

ENDOCRINE THERAPY

Mechanism of action

- Induces apoptosis and exerts cytotoxic behaviour towards leukemia, myeloma, and lymphoma cells

Common side effects

- Gastritis
- Hypertension
- Diabetes
- Fluid retention
- Skin changes
- Insomnia
- Adrenal insufficiency
- Infections

Examples

- Prednisone
- Dexamethasone

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BIOLOGICAL RESPONSE MODIFIERS

IMMUNOTHERAPY

Mechanism of action	• Stimulates body's immune response • May have direct anti-tumor activities	
Common side effects	• Flu-like symptoms • Fever • Malaise/fatigue • Rigors/chills	• Headache • Myelosuppression • Nausea • Rash
Examples	• Interferon • Interleukin-2	

IMMUNE CHECKPOINT INHIBITORS

IMMUNOTHERAPY

Mechanism of action

- Cancer cells utilize checkpoints of the immune system to avoid being killed by the body's natural immune response
- Immune checkpoint inhibitors block checkpoint proteins, boosting the immune response so cancer cells can be identified and killed

Common side effects

- Immune-mediated adverse effects
 - Most common: skin rash, colitis, arthralgia, hepatitis, pneumonitis, endocrine (e.g. thyroid) toxicities
 - Can affect any organ system in the body

Examples

- Pembrolizumab, nivolumab, ipilimumab, cemiplimab

IMMUNOMODULATORY DRUGS

IMMUNOTHERAPY

Mechanism of action	- Stimulate the immune system to attack cancer cells (exact mechanism unknown) - These drugs may also reduce angiogenesis, inflammation, and cancer cell proliferation
Common side effects	- Rash and/or pruritus - Fatigue - Myelosuppression - Diarrhea or constipation - Thromboembolic events - Hepatotoxicity
Examples	Lenalidomide, pomalidomide, thalidomide

MONITORING PARAMETERS

Efficacy

- Response to therapy
 - Remission
 - Symptom improvement
 - Cure

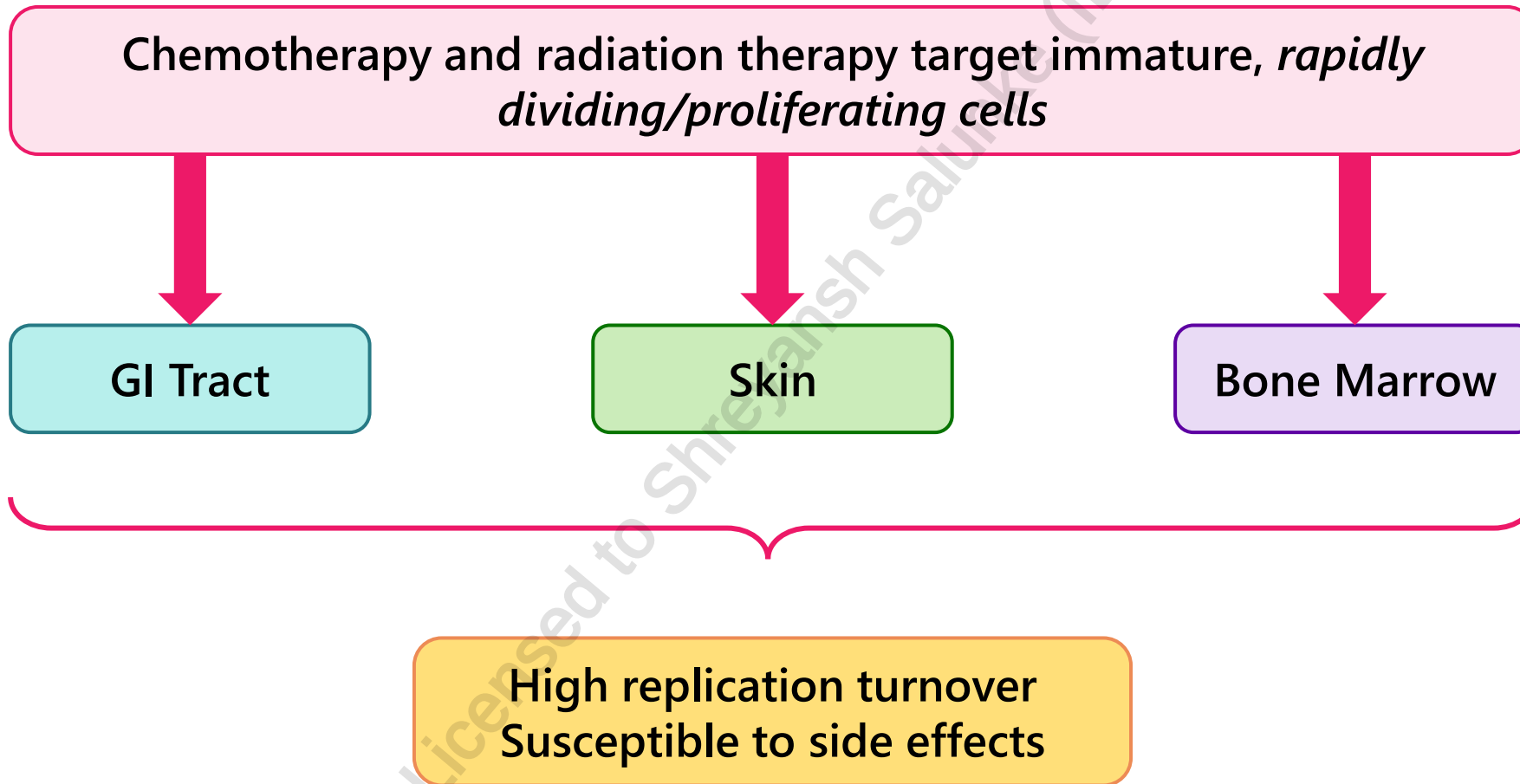
Toxicity

- Side effects
 - Most prevalent in sites with high cellular turnover

Drug Interactions

- Micromedex®
- UpToDate/UpToDate® Lexidrug
- Facts & Comparisons®

MANAGING CHEMOTHERAPY TOXICITY



UPPER GI TRACT TOXICITY

Radiation can cause fibrosis of the salivary glands, muscles, taste buds, and vasculature, leading to mucosal atrophy

Chemotherapy can reduce the rate of renewal for mucosal tissue (Normally 7-14 days)

Mucositis (Typically appears at least **5-7 days after chemotherapy**)

- Maintain good oral hygiene, e.g. brushing, flossing, oral rinses
- Cryotherapy (e.g. ice chips swished around)
- Treat secondary infections if they arise
- Administer topical or systemic pain medication, examples:
 - **Benzydamine** – locally acting NSAID
 - **Magic Mouthwash** – swish and spit (local action)
- Avoid alcohol and excess sugar

Xerostomia (dry mouth)

- Only **symptomatic** relief is available
 - Sugar free gum, Biotene® toothpaste, oral rinses

DIARRHEA

- Certain chemotherapy medications (**alkylating agents, antimetabolites, taxanes, platinum, irinotecan**, etc.) have high rates of severe diarrhea
 - Severe irritation of the lower GI tract, which kills mucosal cells and results in forced evacuation

Treatment:

- **Loperamide**
 - Normal dose:
 - 4 mg STAT, then 2 mg with every loose bowel movement (max 16 mg/day)
 - High dose (for chemotherapies that are prone to cause diarrhea):
 - 4 mg STAT, then 2 mg Q2H until 12 hours after last loose bowel movement
- If loperamide not effective:
 - Can use Lomotil® (**diphenoxylate/atropine**) or Sandostatin® (**octreotide**)

Always maintain adequate hydration! If unable to do this, a referral to the Emergency Department is warranted.

ALOPECIA

- Chemotherapy can cause alopecia by damaging hair follicles responsible for growth
- Patients may experience total or patchy hair loss; some patients may only experience changes in hair quality (thinning, dulling, drying of hair)
- Hair loss may be experienced in all areas of the body including the scalp, face (eyelashes, eyebrows, and beard), arms, or legs
- Taxanes, doxorubicin, cyclophosphamide, and vincristine are commonly associated with alopecia
- Hair loss typically starts 1-3 weeks after first chemotherapy treatment
- Hair typically starts to grow back 6-8 weeks after last treatment
- Cold caps (scalp cooling) can help reduce hair loss in some cases. Hair and scalp care during and after treatment includes:
 - Choosing milder shampoos
 - Using softer hairbrushes
 - Protecting the scalp from sunlight
 - Using cosmetic wigs or caps

EGFR RASH

- Inhibition of tyrosine kinase pathway of keratinocytes occurs in 50% of patients on EGFR inhibitors within the first 1-2 weeks
- Presents as erythema, edema, papulopustular eruptions followed by crusting and dryness of the skin
- Treatment:
 - Moisturizing creams
 - Topical or systemic steroids
 - Antihistamines
 - Topical/systemic antibiotics

PAIN

Pain is a common symptom of cancer and clinicians should always include the patient in the decision-making process for pain

MILD PAIN

- Non-opioid analgesics (acetaminophen, ibuprofen)

MODERATE PAIN

- Weaker opioids (codeine, oxycodone, hydromorphone, morphine)

Moderate-Severe

- Higher doses of morphine, fentanyl, hydromorphone



Consider therapy augmentation with gabapentin, pregabalin, TCAs, SNRIs

MYELOSUPPRESSION AND ANEMIA

- Cytotoxic agents affect bone marrow cells
 - Rapid self-renewal and differentiation is impeded
 - Megakaryocytes, erythrocytes, granulocytes reduced
 - Differentiation normally takes 10-14 days
 - Subsequent neutropenia and anemia
 - Potential for nadir (lowest point)
 - Typically occurs 7-10 days after administration of chemotherapy

NEUTROPENIA (LOW NEUTROPHIL COUNT)

Prevention

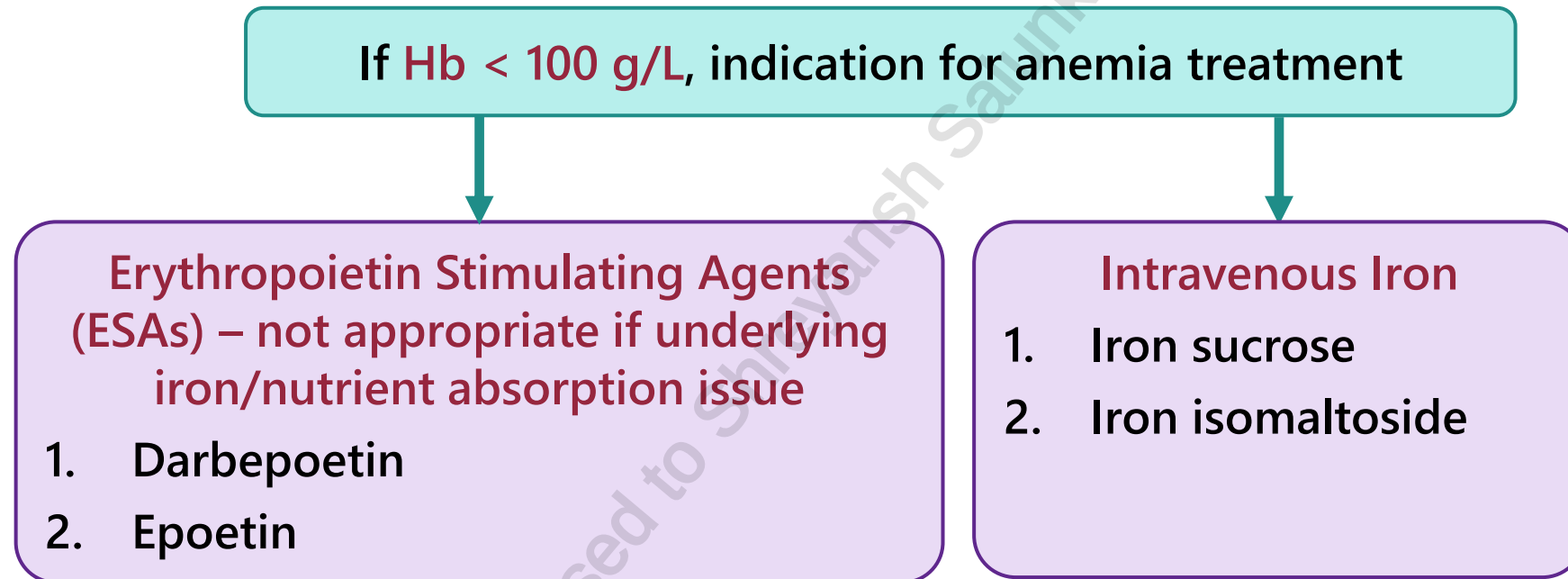
- Lifestyle infection prevention (hand washing)
- Maintaining updated immunizations (**inactivated vaccines only**)
 - Live vaccines should be given >4 weeks before and not until 3 to 6 months after treatment
- Decrease chemotherapy dose (if previous issues with neutropenia)
- Medication: Granulocyte Colony Stimulating Factors (G-CSF) if risk of neutropenia >20%
 - **Filgrastim x 7 days**
 - **Pegfilgrastim x 1 dose**

Treatment

- Report fever immediately during neutropenia
- Initiate antibiotics if FEBRILE NEUTROPENIA – draw 2 sets of blood cultures first
- Use broad-spectrum coverage
 - Example: Piperacillin-tazobactam

ANEMIA

Cytotoxic agents suppress erythrocyte maturation. However, they have a lifespan of 120 days and anemia is therefore not noticed until later after chemotherapy has started (usually noticed after multiple chemotherapy cycles)



Not everyone with anemia is a candidate for ESA!

OTHER SIDE EFFECTS

Coronary Vasospasm/Myocardial Ischemia: Occurs with fluoropyrimidines such as capecitabine or fluorouracil (5-FU)

Prevention

- Use with caution in patients with stable coronary artery disease (CAD)
- Avoid in patients with severe or unstable CAD
- Use of calcium-channel blockers (CCBs) may be beneficial for patients at risk

Management

- **Reversible** after stopping drug
- **Start CCBs + nitrates for symptoms**

OTHER SIDE EFFECTS

Heart Failure

When it occurs with **lapatinib** and **trastuzumab**:

Prevention

- Cardiac monitoring while the patient receives the drug and more frequent monitoring of EF if cardiac risk factors

Management

- **Discontinue drug if patient is symptomatic or interrupt if patient is asymptomatic with poor LV function**
- Standard heart failure management

When it occurs with **anthracyclines**:

Prevention

- Limit doses to suggested maximum cumulative doses
- Monitor LV function
- **Dexrazoxane** for patients whose lifetime cumulative dose exceeds safe limits and those with underlying cardiac risk factors

Management

- **Re-evaluate risks and benefits of continuation with medical team and patient**
- Standard heart failure management

OTHER SIDE EFFECTS

Thromboembolism: occurs with **angiogenesis inhibitors** (bevacizumab, sunitinib) and **thalidomide**

Prevention

- Provide DVT prophylaxis in patients receiving thalidomide in the ambulatory setting
 - Oncologist assessment first

Management

- Use a **LMWH or DOAC**. LMWH preferred if cancer-associated VTE and have one or more of:
 - Malignancy of GI/GU tract or GI/GU bleeding
 - Thrombocytopenia (platelet count 25-50x10⁹/L)
 - Pregnant or breastfeeding
 - Treating immediately for VTE with large clot
- **Avoid warfarin in patients with cancer**

Duration of Therapy

Treat for at least 3 to 6 months. Consider treating for longer if meets any of the following:

- Continuing to receive systemic chemotherapy
- Has metastatic disease
- Has progressive or relapsed disease
- Has ongoing risk factors that increase the risk of recurrent thrombosis
- Has low bleeding risk

OTHER SIDE EFFECTS

Extravasation: occurs with anthracyclines, antitumour antibiotics, taxanes, vinca alkaloids, platinum compounds, and mechlorethamine

Prevention

- IV lines should be tested for good flow
- Place IV line appropriately and monitor site during infusion

Management

- Stop infusion and aspirate drug from tissue
- Apply cold compresses for 20 minutes four times daily for 1 to 2 days
- Local heat application should be used for vinca alkaloids
- Antidotes
 - Dimethyl sulfoxide (DMSO) for anthracyclines and antitumour antibiotics
 - Dexrazoxane for anthracyclines
 - Sodium thiosulfate for cisplatin

OTHER SIDE EFFECTS

Hand-foot syndrome (palmar-plantar erythrodysesthesia): occurs with **capecitabine**, **liposomal doxorubicin**, **sorafenib**, **sunitinib**, **fluorouracil**

Prevention

- **Apply moisturizers REGULARLY**
- Cool hands and feet frequently
- Avoid ill-fitting shoes and clothing
- Avoid hot water and use gloves when handwashing dishes

Management

- **Dose interruption and decrease**
- Discontinue the drug if severe or if symptoms recur even with a decreased dose
- Emollient applied on ulcerations and desquamation

OTHER SIDE EFFECTS

Infusion Reactions: occurs with **monoclonal antibodies**

Prevention

- Consider pre-treatment with corticosteroid and antihistamine (may not be required for humanized monoclonal antibodies – trastuzumab, bevacizumab, panitumumab)
- May use dexamethasone, hydrocortisone, diphenhydramine, and famotidine amongst others

Management

- A severe reaction is defined by a rash covering >30% BSA (body surface area), hypoxia, edema, angioedema, dizziness, hypotension (>20mmHg drop from baseline), and fever >40 degrees Celsius
 - In severe cases, the infusion should be stopped, and the patient should be given a corticosteroid, bronchodilator, IV fluids, and an antihistamine. The causative agent should be discontinued
- A mild-moderate case is marked by mild flushing, chills, dizziness not interfering with activity, pruritus, transient rash (<30% BSA)
 - Treat and restart infusion at a slower rate, give pre-treatment medication for continuing and next infusions

OTHER SIDE EFFECTS

Tumor Lysis Syndrome: occurs with large tumor burden

Prevention

- Aggressive hydration and diuretics to maintain good urine output
- Rasburicase, allopurinol or febuxostat to prevent hyperuricemia

Management

- Treat hyperkalemia, hyperphosphatemia, and hypocalcemia if symptomatic or severe
- Start dialysis if acute renal failure
- Rasburicase

OTHER SIDE EFFECTS

Hemorrhagic Cystitis: occurs with **cyclophosphamide** and **ifosfamide**

Prevention

- **Mesna** is a compound infused before chemotherapy that helps inactivate acrolein, the compound associated with hemorrhagic cystitis
- **Diuresis and hydration while on chemotherapy**
- Give chemotherapy **early in the day to ensure adequate urination** post-chemotherapy to avoid prolonged contact with damaging metabolite
- Educate the patient about hydration post-chemotherapy

Management

- Bladder irrigation, intravesical astringents, or systemic antifibrinolytics for mild cases
- Cystoscopy or surgery for severe cases

OTHER SIDE EFFECTS

Hot Flashes: occurs with **anti-hormonal agents**

Prevention

- Avoid triggers: caffeine, hot temperature, stress

Management

- **Venlafaxine**, clonidine, gabapentin
- Short course of estrogen (if estrogen is not contraindicated)
- Exercise and behavioural & relaxation techniques

TAKEAWAY

SUMMARY OF MAIN CHEMOTHERAPY-INDUCED SIDE EFFECTS

Cardiac vasospasms:
capecitabine or 5-fluorouracil

Heart failure:
lapatinib, trastuzumab, anthracyclines

VTE:
bevacizumab, sunitinib, thalidomide

Extravasation

Hand-foot syndrome:
capecitabine, doxorubicin, sorafenib, sunitinib, 5-fluorouracil

CASE 1

DS is a 58-year-old male with a recent diagnosis of colorectal cancer. He presented with a 3-month history of decreased oral intake and hematochezia. A colonoscopy found a large mass in the colon, and a biopsy confirmed colorectal cancer. One week ago, DS received his first cycle of 5-fluorouracil and irinotecan (FOLFIRI) with palliative intent. Today, DS reports he is experiencing 5 loose bowel movements per day. He has a past medical history of type 2 diabetes and neuropathy. He takes metformin 1000 mg PO BID, sitagliptin 100 mg PO daily, empagliflozin 10 mg PO daily, pregabalin 75 mg PO BID, ginseng, vitamin C 500 mg, and a multivitamin daily. He has a penicillin allergy resulting in a rash. He smokes ½ pack of cigarettes per day and has a 20-pack-year history of smoking.

1. Which of the following is the most likely cause of DS' diarrhea?
 - a) Metformin
 - b) Colorectal cancer
 - c) Irinotecan
 - d) *Clostridium difficile* infection



CASE 1

DS is a 58-year-old male with a recent diagnosis of colorectal cancer. He presented with a 3-month history of decreased oral intake and hematochezia. A colonoscopy found a large mass in the colon, and a biopsy confirmed colorectal cancer. One week ago, DS received his first cycle of 5-fluorouracil and irinotecan (FOLFIRI) with palliative intent. Today, DS reports he is experiencing 5 loose bowel movements per day. He has a past medical history of type 2 diabetes and neuropathy. He takes metformin 1000 mg PO BID, sitagliptin 100 mg PO daily, empagliflozin 10 mg PO daily, pregabalin 75 mg PO BID, ginseng, vitamin C 500 mg, and a multivitamin daily. He has a penicillin allergy resulting in a rash. He smokes ½ pack of cigarettes per day and has a 20-pack-year history of smoking.

2. Which medication is first line for treating DS' diarrhea?
- a) Vancomycin
 - b) Loperamide
 - c) Diphenoxylate/atropine
 - d) Diphenoxylate/atropine and Loperamide



CASE 1

DS is a 58-year-old male with a recent diagnosis of colorectal cancer. He presented with a 3-month history of decreased oral intake and hematochezia. A colonoscopy found a large mass in the colon, and a biopsy confirmed colorectal cancer. One week ago, DS received his first cycle of 5-fluorouracil and irinotecan (FOLFIRI) with palliative intent. Today, DS reports he is experiencing 5 loose bowel movements per day. He has a past medical history of type 2 diabetes and neuropathy. He takes metformin 1000 mg PO BID, sitagliptin 100 mg PO daily, empagliflozin 10 mg PO daily, pregabalin 75 mg PO BID, ginseng, vitamin C 500 mg, and a multivitamin daily. He has a penicillin allergy resulting in a rash. He smokes ½ pack of cigarettes per day and has a 20-pack-year history of smoking.

3. How is loperamide dosed for chemotherapy-associated diarrhea?
- a) 4mg PO STAT, then 4 mg PO after every loose BM
 - b) 2mg PO STAT, then 2mg PO after every loose BM
 - c) 4mg PO STAT, then 2mg PO Q2H until 12 hours after last loose bowel movement
 - d) 4mg PO STAT, then 4mg PO Q2H until 12 hours after last loose bowel movement



CASE 2

DS is a 58-year-old male with a recent diagnosis of colorectal cancer. He presented with a 3-month history of decreased oral intake and hematochezia. A colonoscopy found a large mass in the colon and a biopsy confirmed colorectal cancer. One week ago, DS received his first cycle of 5-fluorouracil and irinotecan (FOLFIRI) with palliative intent. Today, while receiving his infusion, DS mentioned that his hands and feet are painful and red. He has a past medical history of type 2 diabetes and neuropathy. He takes metformin 1000 mg PO BID, sitagliptin 100 mg PO daily, empagliflozin 10 mg PO daily, pregabalin 75 mg PO BID, ginseng, vitamin C 500 mg, and a multivitamin daily. He has a penicillin allergy resulting in a rash. He smokes ½ pack of cigarettes per day and has a 20-pack-year history of smoking.



CASE 2

1. Which strategy below is NOT used to avoid hand-foot syndrome?

- a) Apply moisturizer regularly
- b) Avoid tight footwear
- c) Use gloves when washing dishes
- d) Apply heat to hands and feet following chemotherapy

2. What are reasonable options to treat DS' hand-foot syndrome?

- a) Avoid applying anything to the area and stop the chemotherapy
- b) Change the current chemotherapy regimen
- c) Delay chemotherapy and increase doses for future cycles
- d) Delay chemotherapy and reduce doses for future cycles

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CHANGE LOG

August 2025

- Minor formatting changes made
- Slide 5: Minor rewording of N0
- Slide 12: Removed 'teniposide' as no longer available on the Canadian market
- Slide 15: Removed 'streptozocin' as no longer available on the Canadian market

October 2025

- No content changes made

March 2026

- Slide 2: Updated NAPRA competencies
- Formatting changes made throughout
- Slides 10, 18, 22, 28: Corrected typo – SERM stands for selective estrogen receptor modulator
- Slide 26: Removed nilutamide as not on Canadian market anymore
- Slide 34: Added cryotherapy
- Slide 36: Minor updates
- Slide 41: Minor rephrasing for clarity
- Slide 57: Updated references